

Capital Reserve Depletion and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Reserve Depletion (CRD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CRD achieves an annualized gross (net) Sharpe ratio of 0.54 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (20) bps/month with a t-statistic of 3.08 (2.99), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Net equity financing) is 16 bps/month with a t-statistic of 2.45.

1 Introduction

Financial markets continuously process vast amounts of information, yet mounting evidence suggests that prices do not instantaneously reflect all available data. The speed and completeness of information incorporation into asset prices remains a central question in finance, with implications for market efficiency, return predictability, and optimal investment strategies. While classical efficient market theory posits rapid price adjustment to new information, behavioral and institutional frictions create systematic delays in how market participants process and act upon firm-specific signals. This paper examines Capital Reserve Depletion (CRD), a signal derived from changes in firms' capital reserves relative to their stock capital, as a predictor of future stock returns. We argue that CRD captures gradual information diffusion about firms' financial health and capital management decisions that investors process slowly due to limited attention and cognitive constraints.

The slow diffusion of information provides a compelling framework for understanding why CRD predicts returns. When firms deplete their capital reserves—through dividend payments, share repurchases, or operational losses—this signals important information about future profitability and financial flexibility that diffuses gradually through the market. Building on the seminal work of [Hong et al. \(2000\)](#) on gradual information flow and [DellaVigna and Pollet \(2009\)](#) on investor inattention, we hypothesize that market participants underreact to capital reserve changes due to limited attention and information processing constraints. The complexity of interpreting capital reserve movements, which requires analyzing multiple financial statement items and understanding their implications for future cash flows, exacerbates these attention limitations.

The gradual incorporation of CRD information should manifest most strongly where information frictions are severe. Following [Hirshleifer and Teoh \(2003\)](#) and [Peng and Xiong \(2007\)](#), we expect the return predictability from CRD to concentrate

among stocks with higher information processing costs—particularly small firms, those with low analyst coverage, and stocks with limited institutional ownership. These firms face greater barriers to information dissemination, as documented by [Hong et al. \(2000\)](#), who show that bad news travels slowly among small stocks, and [Zhang \(2006\)](#), who demonstrates that information uncertainty amplifies underreaction to public signals.

The economic mechanism linking CRD to returns operates through investors’ delayed recognition of deteriorating financial positions. When firms deplete capital reserves without corresponding operational improvements, this signals potential financial distress or suboptimal capital allocation decisions that investors initially overlook. As information gradually diffuses and more sophisticated investors recognize these signals, prices adjust slowly toward fundamental values, generating predictable return patterns. This gradual price adjustment process, consistent with [Cohen and Frazzini \(2008\)](#)’s evidence on slow information diffusion across economically linked firms, creates profitable trading opportunities for investors who identify and act upon CRD signals before the broader market fully incorporates this information.

We find strong evidence that CRD predicts stock returns in the cross-section. A value-weighted long/short trading strategy based on CRD achieves an annualized gross Sharpe ratio of 0.54, placing it in the top 8

The predictive power of CRD extends across the size spectrum, supporting our hypothesis of gradual information diffusion. Among the largest stocks—those above the 80th NYSE percentile—the CRD strategy delivers an average return of 25 basis points per month (t -statistic = 2.94), with alphas ranging from 21 to 28 basis points across standard factor models. This persistence among large, liquid stocks suggests that the CRD effect reflects systematic underreaction rather than market microstructure effects or limits to arbitrage. The strategy maintains significant al-

phas even when using alternative sorting methods, including equal-weighting and different breakpoint choices, with twenty-three out of twenty-five specifications yielding t-statistics exceeding three.

Controlling for related signals strengthens our interpretation of CRD as capturing distinct information about gradual capital structure deterioration. When we control for the six most closely related anomalies—including share issuance measures and payout yields—plus the six Fama-French factors, the CRD strategy retains an alpha of 16 basis points per month (t-statistic = 2.45). This residual predictability indicates that CRD captures unique information about firms’ financial health not subsumed by existing predictors. Furthermore, the strategy’s net Sharpe ratio of 0.47 ranks in the top 2

This paper contributes to the literature on information diffusion and cross-sectional return predictability by documenting a novel signal based on capital reserve depletion that captures gradual market learning about firms’ financial positions. Our work extends [Cohen et al. \(2003\)](#)’s findings on post-earnings announcement drift and [Sloan \(1996\)](#)’s accrual anomaly by identifying a balance sheet signal that predicts returns through systematic underreaction rather than earnings management or accounting distortions. Unlike [Daniel and Titman \(2006\)](#), who focus on investor overconfidence and biased self-attribution, we demonstrate that limited attention to complex financial signals drives predictable return patterns even among large, well-analyzed firms.

Methodologically, we advance the anomaly literature by rigorously testing CRD using the protocol of [Novy-Marx and Velikov \(2023\)](#), ensuring our results meet the highest standards for robustness and replicability. Our comprehensive analysis spans multiple portfolio construction methods, controls for transaction costs using high-frequency effective spreads from [Chen and Velikov \(2022\)](#), and examines performance

relative to 212 documented anomalies. This systematic approach addresses concerns raised by [Harvey et al. \(2016\)](#) about data mining and [McLean and Pontiff \(2016\)](#) about post-publication decay, demonstrating that CRD represents a genuine source of alpha rather than a statistical artifact.

Our findings have important implications for market efficiency and investment practice. The persistence of CRD’s predictive power across size groups and after controlling for known factors challenges strong-form efficiency while supporting models of gradual information diffusion and limited investor attention. For practitioners, CRD offers an implementable strategy that generates significant risk-adjusted returns net of transaction costs, expanding the investable opportunity set beyond existing factors. More broadly, our results suggest that complex balance sheet signals remain underexploited sources of alpha, as investors struggle to process and incorporate multifaceted financial information in real-time, creating opportunities for sophisticated investors who can identify and act upon these gradual information flows.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of capital structure changes across firms. Capital Reserve Depletion signal relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding, reflecting the nominal value assigned to issued common equity on the balance sheet. Capital surplus (CAPS), also known as additional paid-in capital, captures the amount shareholders have invested in the firm above the par value of common stock, typically arising from

stock issuances at prices exceeding par value. Together, these components constitute important elements of shareholders' equity, with changes in their relationship potentially signaling shifts in a firm's equity financing activities or capital management strategies. We construct the Capital Reserve Depletion signal by computing the year-over-year change in common stock scaled by lagged capital surplus: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{CAPS}(t-1)$. This ratio captures the rate at which firms alter their common stock relative to their existing capital surplus base, potentially indicating equity restructuring activities, stock splits, or changes in par value assignments that may affect the composition of shareholders' equity. We calculate this measure using fiscal year-end values to ensure temporal consistency across firms with different fiscal year endings. To maintain data quality, we require non-missing values for both CSTK and CAPS variables, and exclude observations where the lagged capital surplus equals zero to avoid undefined ratios. Firms with negative capital surplus values are retained in the sample, as these may convey meaningful information about capital structure distress or unusual equity transactions.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CRD signal. Panel A plots the time-series of the mean, median, and interquartile range for CRD. On average, the cross-sectional mean (median) CRD is -0.06 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CRD data. The signal's interquartile range spans -0.14 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the CRD signal for the CRSP universe. On average, the CRD signal is available for 5.78% of CRSP names, which on average make up 6.74% of total market capitalization.

4 Does CRD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CRD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CRD portfolio and sells the low CRD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CRD strategy earns an average return of 0.29% per month with a t-statistic of 4.23. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.21% to 0.32% per month and have t-statistics exceeding 3.08 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.26, with a t-statistic of 5.64 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 531 stocks and an average market capitalization of at least \$1,363 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.53. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-32bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 3.00. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CRD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the CRD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CRD, as well as average returns

and alphas for long/short trading CRD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CRD strategy achieves an average return of 25 bps/month with a t-statistic of 2.94. Among these large cap stocks, the alphas for the CRD strategy relative to the five most common factor models range from 21 to 28 bps/month with t-statistics between 2.38 and 3.32.

5 How does CRD perform relative to the zoo?

Figure 2 puts the performance of CRD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CRD strategy falls in the distribution. The CRD strategy’s gross (net) Sharpe ratio of 0.54 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CRD strategy (red line).² Ignoring trading costs, a \$1 invested in the CRD strategy would have yielded \$6.31 which ranks the CRD strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CRD strategy would have yielded \$4.49 which ranks the CRD strategy in the top 2% across the 212 anomalies.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CRD relative to those. Panel A shows that the CRD strategy gross alphas fall between the 63 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CRD strategy has a positive net generalized alpha for five out of the five factor models. In these cases CRD ranks between the 80 and 86 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CRD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CRD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CRD or at least to weaken

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

the power CRD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CRD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRD}CRD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CRD. Stocks are finally grouped into five CRD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CRD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CRD signal in these Fama-MacBeth regressions exceed -0.05, with the minimum t-statistic occurring when controlling for Share issuance (5 year). Controlling for all six closely related anomalies, the t-statistic on CRD is 0.24.

Similarly, Table 5 reports results from spanning tests that regress returns to the CRD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CRD strategy earns alphas that range from 13-24bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 2.02, which is achieved when controlling for Share issuance (5 year). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CRD trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.45.

7 Does CRD add relative to the whole zoo?

Finally, we can ask how much adding CRD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CRD signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CRD grows to \$2389.63.

8 Conclusion

This study provides compelling evidence that Capital Reserve Depletion (CRD) represents a economically meaningful and statistically robust predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CRD is available.

Novy-Marx and Velikov (2023), demonstrates that CRD-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CRD delivers an annualized Sharpe ratio of 0.54 gross (0.47 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. The strategy’s ability to generate monthly abnormal returns of 21 basis points gross relative to the Fama-French five-factor model plus momentum, with a highly significant t-statistic of 3.08, underscores its robustness as a return predictor. Particularly noteworthy is the signal’s resilience when confronted with related equity financing and payout measures—the strategy maintains a statistically significant alpha of 16 basis points per month (t-statistic = 2.45) even after controlling for six closely related factors including share issuance metrics, book equity growth, and net payout yield.

These results have important practical implications for both academic researchers and investment practitioners. For academics, CRD adds to our understanding of how firms’ capital management decisions relate to expected returns, potentially reflecting either risk-based explanations or market inefficiencies in processing complex financial information. For practitioners, the signal’s robust performance net of transaction costs and its incremental predictive power beyond existing factors suggest potential value in incorporating CRD into quantitative investment strategies.

Several limitations of this study warrant consideration. First, while we employ conservative transaction cost estimates, actual implementation costs may vary depending on market conditions and execution capabilities. Second, the analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, our study does not definitively resolve whether the CRD premium reflects compensation for systematic risk or

market mispricing, though the persistence of abnormal returns relative to standard risk models suggests the latter may play a role.

Future research should explore several promising avenues. Investigation into the economic mechanisms underlying the CRD effect would enhance our theoretical understanding of this anomaly. Cross-country studies could establish whether the signal's predictive power extends beyond U.S. markets, potentially revealing important institutional or regulatory influences. Additionally, examining the interaction between CRD and corporate events such as mergers, acquisitions, or financial distress could provide insights into when the signal is most informative. Finally, exploring the time-varying nature of the CRD premium and its relationship to macroeconomic conditions could offer valuable insights for dynamic portfolio management strategies. As financial markets continue to evolve, understanding signals like CRD that capture firms' capital allocation decisions remains crucial for both advancing asset pricing theory and improving investment practice.

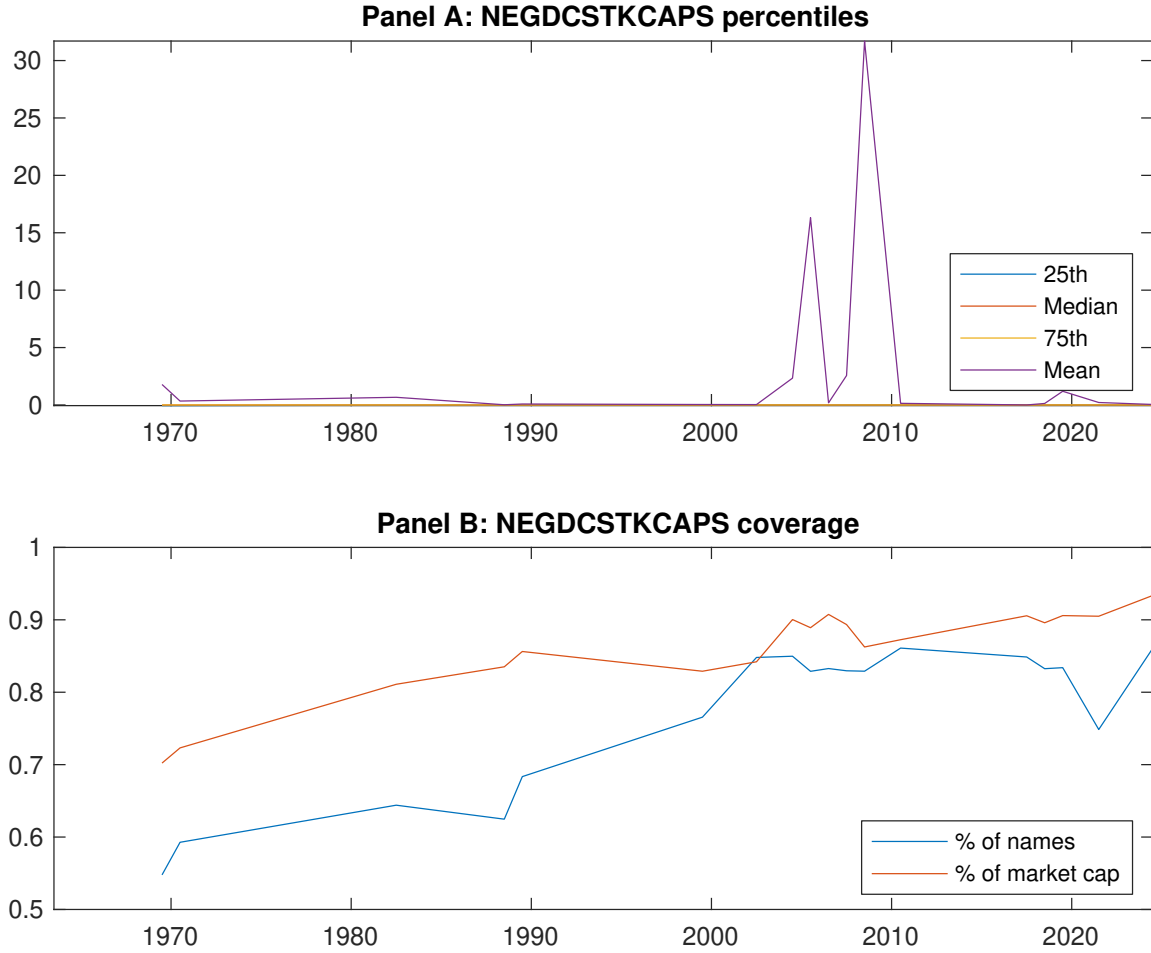


Figure 1: Times series of CRD percentiles and coverage.
This figure plots descriptive statistics for CRD. Panel A shows cross-sectional percentiles of CRD over the sample. Panel B plots the monthly coverage of CRD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CRD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CRD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.59]	0.49 [2.65]	0.62 [3.35]	0.70 [4.25]	0.73 [4.50]	0.29 [4.23]
α_{CAPM}	-0.14 [-3.13]	-0.14 [-2.75]	-0.01 [-0.13]	0.14 [2.84]	0.18 [3.69]	0.32 [4.78]
α_{FF3}	-0.17 [-3.83]	-0.13 [-2.41]	0.02 [0.39]	0.11 [2.29]	0.14 [3.03]	0.31 [4.63]
α_{FF4}	-0.13 [-2.86]	-0.07 [-1.28]	0.03 [0.49]	0.08 [1.57]	0.14 [2.91]	0.27 [3.90]
α_{FF5}	-0.18 [-3.87]	-0.04 [-0.85]	0.05 [0.96]	0.07 [1.44]	0.06 [1.39]	0.24 [3.57]
α_{FF6}	-0.14 [-3.08]	-0.00 [-0.00]	0.06 [1.01]	0.05 [0.91]	0.07 [1.48]	0.21 [3.08]
Panel B: Fama and French (2018) 6-factor model loadings for CRD-sorted portfolios						
β_{MKT}	1.01 [90.64]	1.03 [83.35]	1.03 [75.17]	1.00 [83.95]	0.99 [88.92]	-0.01 [-0.69]
β_{SMB}	-0.01 [-0.53]	-0.01 [-0.54]	0.07 [3.69]	-0.06 [-3.43]	-0.04 [-2.19]	-0.03 [-1.13]
β_{HML}	0.10 [4.50]	-0.01 [-0.42]	-0.12 [-4.69]	0.09 [4.03]	0.03 [1.28]	-0.07 [-2.16]
β_{RMW}	0.07 [3.05]	-0.14 [-5.70]	-0.11 [-4.14]	0.07 [2.83]	0.11 [5.10]	0.05 [1.40]
β_{CMA}	-0.06 [-1.99]	-0.14 [-3.95]	0.04 [1.08]	0.06 [1.69]	0.20 [6.34]	0.26 [5.64]
β_{UMD}	-0.05 [-4.77]	-0.06 [-5.07]	-0.01 [-0.40]	0.04 [3.14]	-0.01 [-0.66]	0.05 [2.76]
Panel C: Average number of firms (n) and market capitalization (me)						
n	531	646	616	638	676	
me (\$10 ⁶)	1475	1363	1857	2170	2324	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CRD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [4.23]	0.32 [4.78]	0.31 [4.63]	0.27 [3.90]	0.24 [3.57]	0.21 [3.08]
Quintile	NYSE	EW	0.48 [7.64]	0.49 [7.86]	0.46 [7.66]	0.40 [6.65]	0.41 [6.87]	0.37 [6.13]
Quintile	Name	VW	0.26 [3.82]	0.30 [4.42]	0.28 [4.18]	0.25 [3.61]	0.22 [3.20]	0.20 [2.84]
Quintile	Cap	VW	0.25 [3.53]	0.28 [3.97]	0.29 [3.96]	0.23 [3.12]	0.25 [3.47]	0.21 [2.84]
Decile	NYSE	VW	0.36 [4.24]	0.38 [4.49]	0.37 [4.29]	0.32 [3.64]	0.36 [4.14]	0.32 [3.64]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [3.65]	0.28 [4.24]	0.27 [4.04]	0.24 [3.64]	0.22 [3.26]	0.20 [2.99]
Quintile	NYSE	EW	0.29 [4.27]	0.32 [4.68]	0.28 [4.28]	0.25 [3.87]	0.23 [3.52]	0.21 [3.23]
Quintile	Name	VW	0.22 [3.25]	0.26 [3.89]	0.24 [3.60]	0.22 [3.29]	0.19 [2.88]	0.18 [2.69]
Quintile	Cap	VW	0.21 [3.00]	0.25 [3.48]	0.24 [3.42]	0.21 [2.96]	0.22 [3.13]	0.20 [2.79]
Decile	NYSE	VW	0.32 [3.73]	0.35 [4.09]	0.33 [3.89]	0.30 [3.55]	0.33 [3.82]	0.30 [3.56]

Table 3: Conditional sort on size and CRD

This table presents results for conditional double sorts on size and CRD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CRD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CRD and short stocks with low CRD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CRD Quintiles					CRD Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.55 [2.33]	0.68 [2.54]	0.80 [3.00]	0.87 [3.45]	0.99 [4.35]	0.47 [6.48]	0.48 [6.60]	0.46 [6.28]	0.40 [5.46]	0.40 [5.30]	0.35 [4.70]
	(2)	0.70 [3.18]	0.62 [2.57]	0.85 [3.51]	0.92 [3.91]	0.95 [4.39]	0.25 [3.03]	0.27 [3.21]	0.24 [2.80]	0.17 [1.97]	0.23 [2.71]	0.17 [2.04]
	(3)	0.60 [2.99]	0.59 [2.69]	0.71 [3.15]	0.92 [4.27]	0.87 [4.45]	0.27 [3.77]	0.29 [4.00]	0.27 [3.77]	0.23 [3.13]	0.25 [3.36]	0.22 [2.87]
	(4)	0.53 [2.82]	0.61 [3.05]	0.73 [3.52]	0.84 [4.34]	0.78 [4.23]	0.25 [3.47]	0.27 [3.71]	0.23 [3.22]	0.23 [3.18]	0.12 [1.71]	0.14 [1.86]
	(5)	0.42 [2.46]	0.52 [2.97]	0.46 [2.48]	0.59 [3.47]	0.67 [4.19]	0.25 [2.94]	0.28 [3.28]	0.28 [3.32]	0.21 [2.48]	0.26 [3.04]	0.21 [2.38]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CRD Quintiles					CRD Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	350	348	347	346	341	31	31	35	27	26	
	(2)	100	97	99	98	97	53	52	53	52	51	
	(3)	71	71	70	71	70	90	89	89	93	91	
	(4)	59	59	59	59	59	188	189	198	197	201	
(5)	53	53	53	53	53	1213	1365	1479	1509	1793		

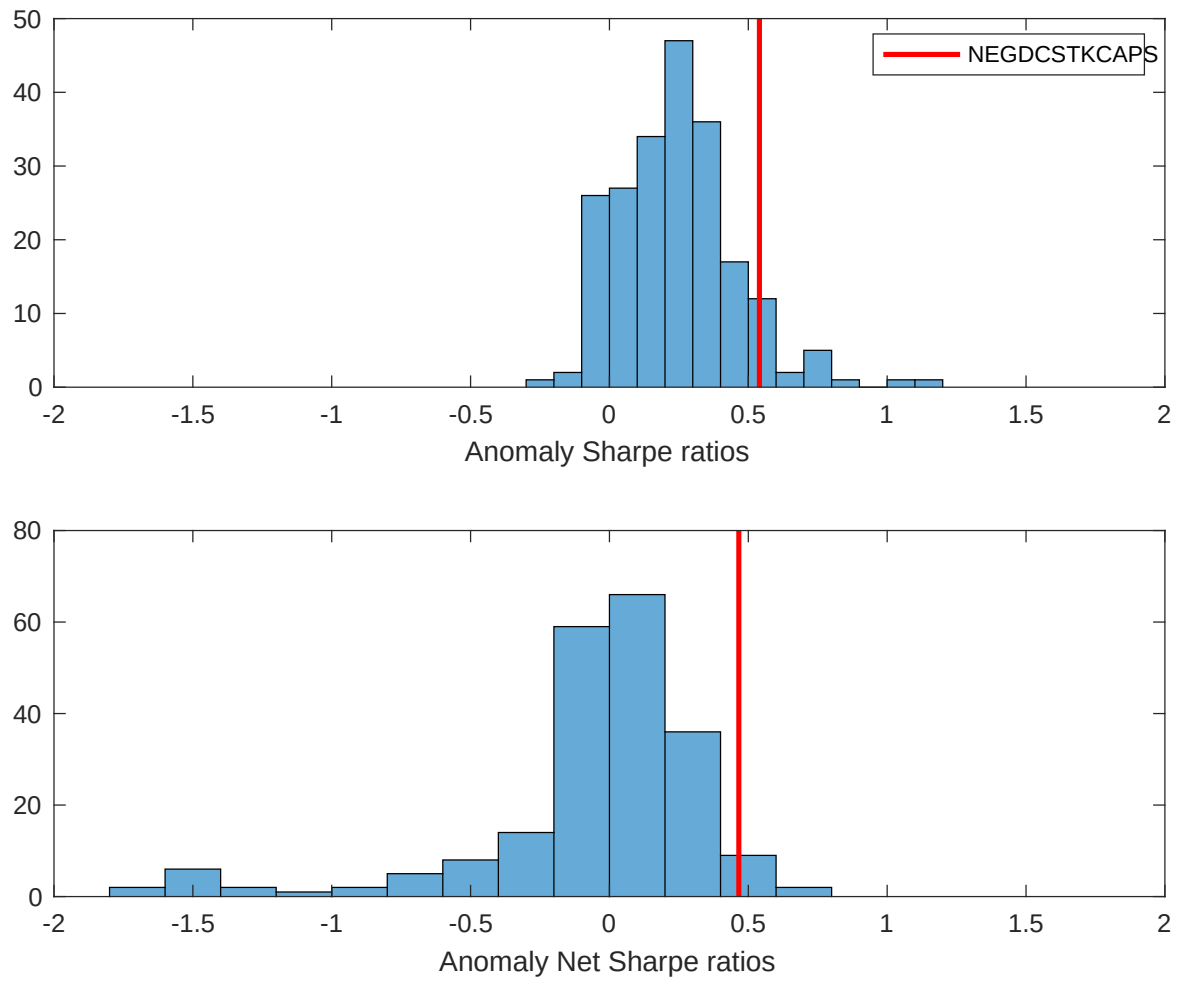


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CRD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

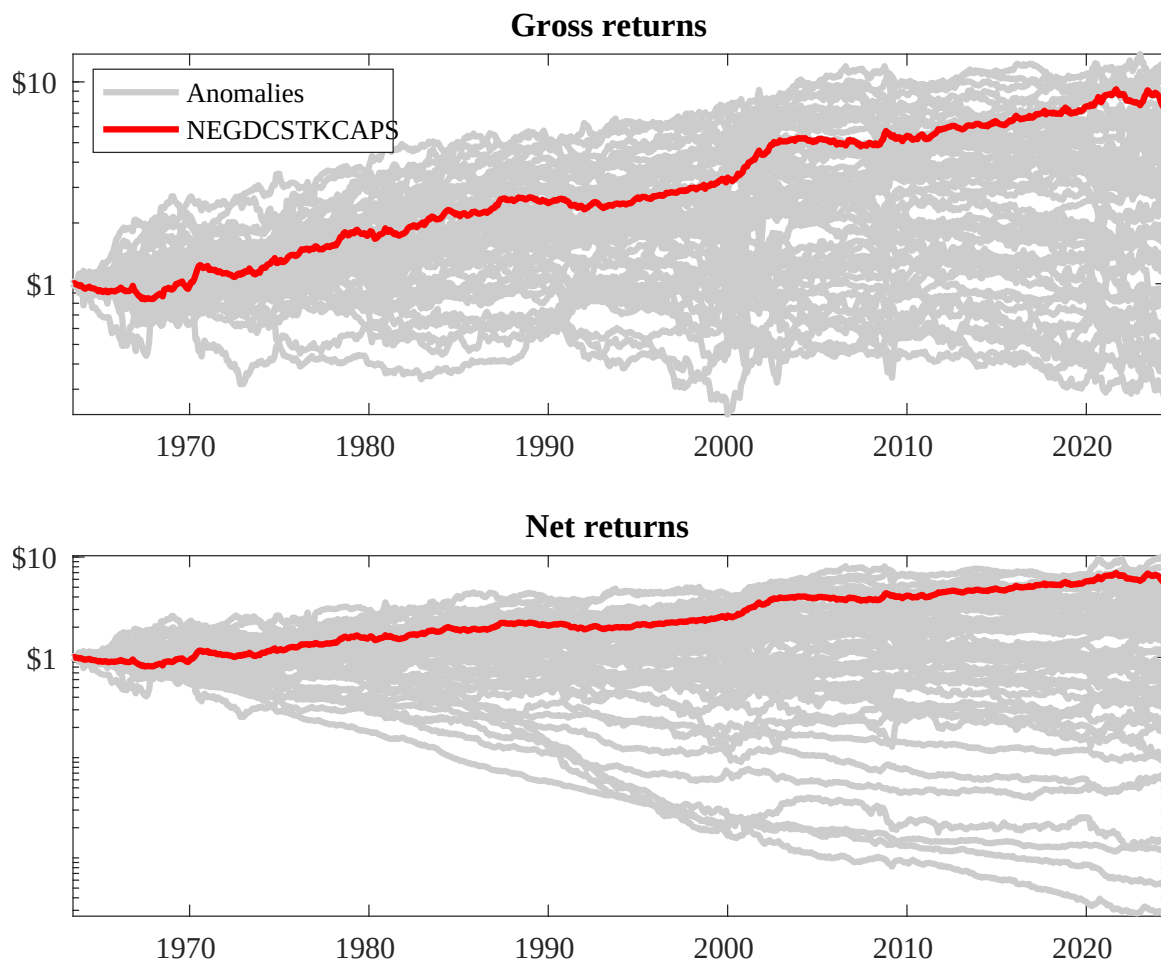
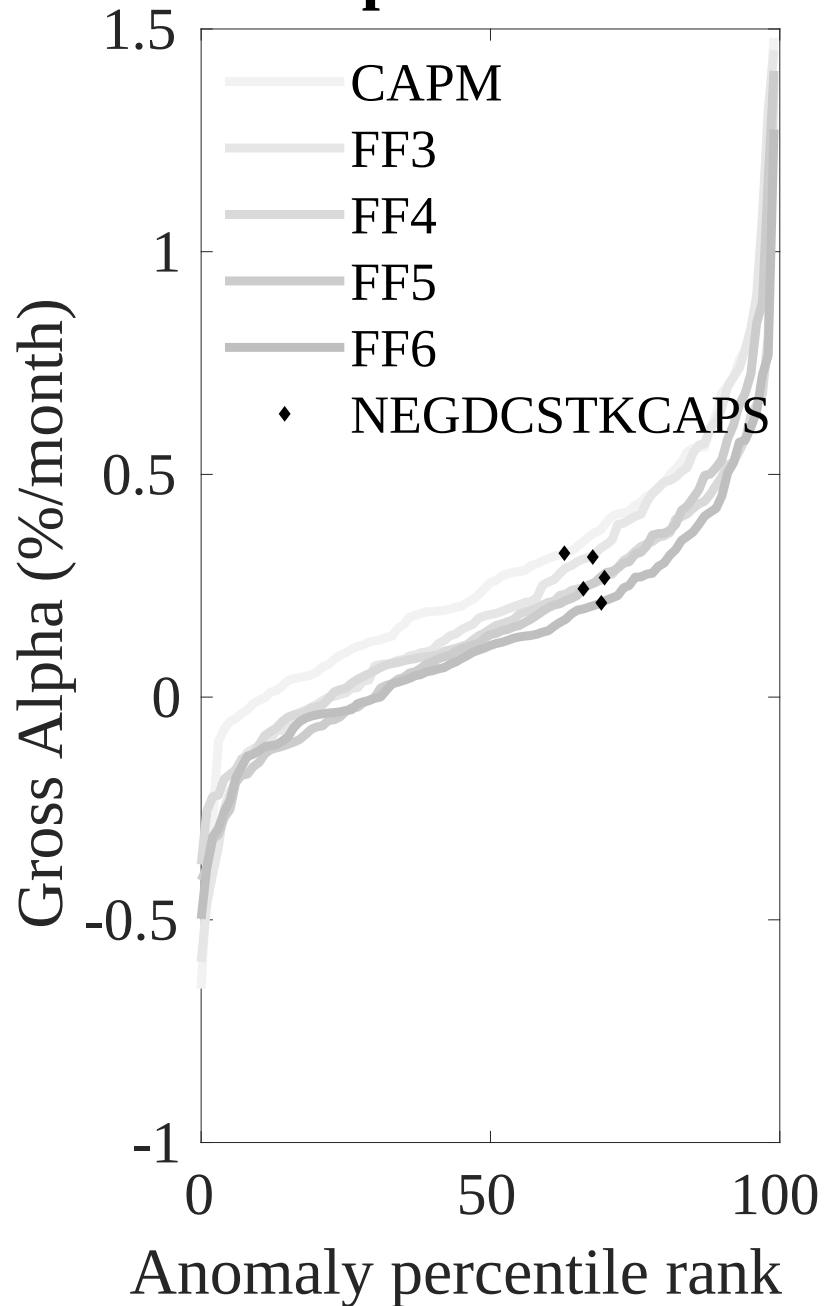


Figure 3: Dollar invested.

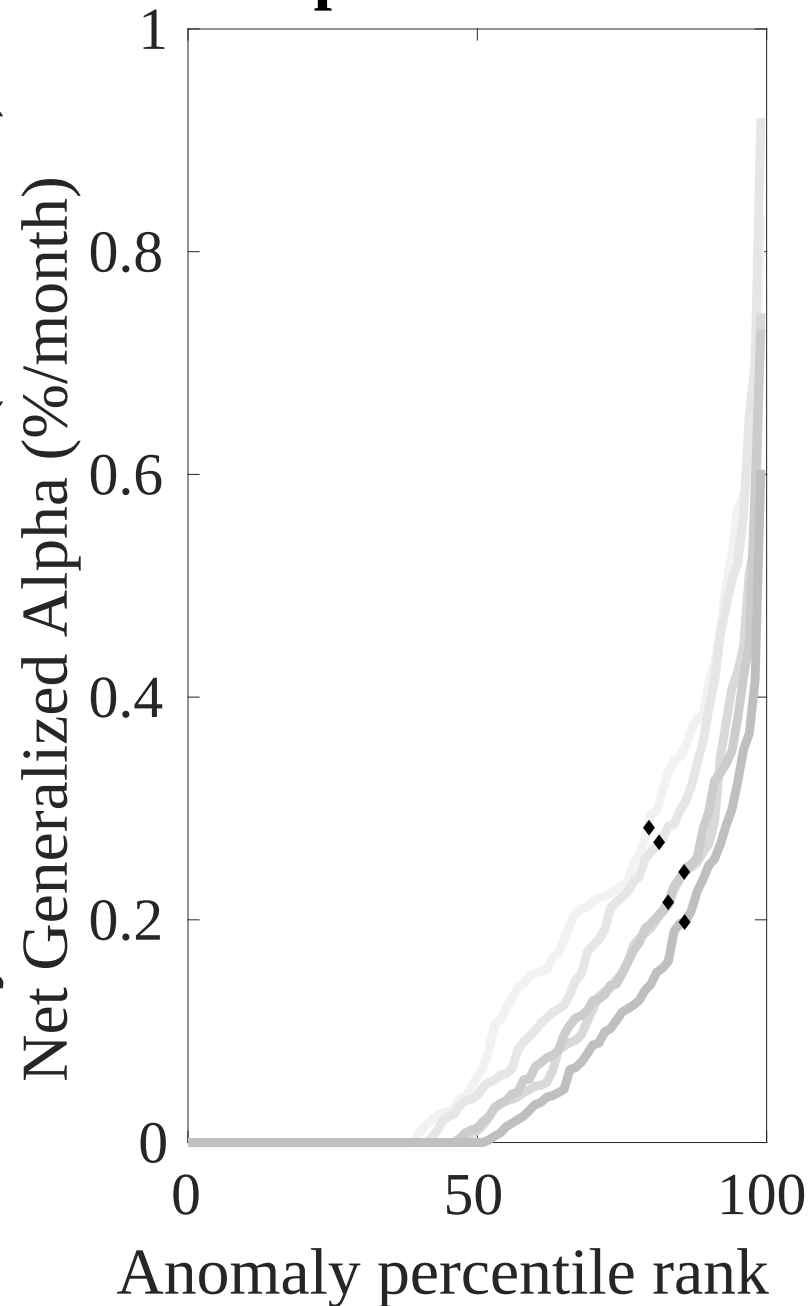
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CRD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



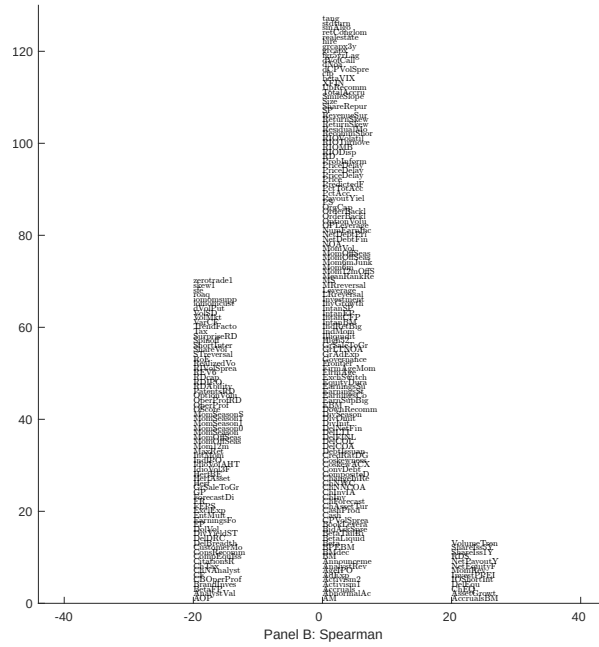
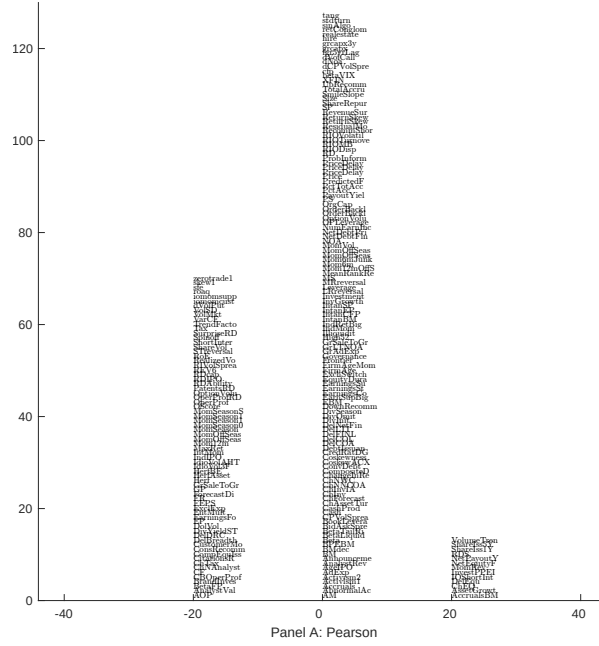


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with CRD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

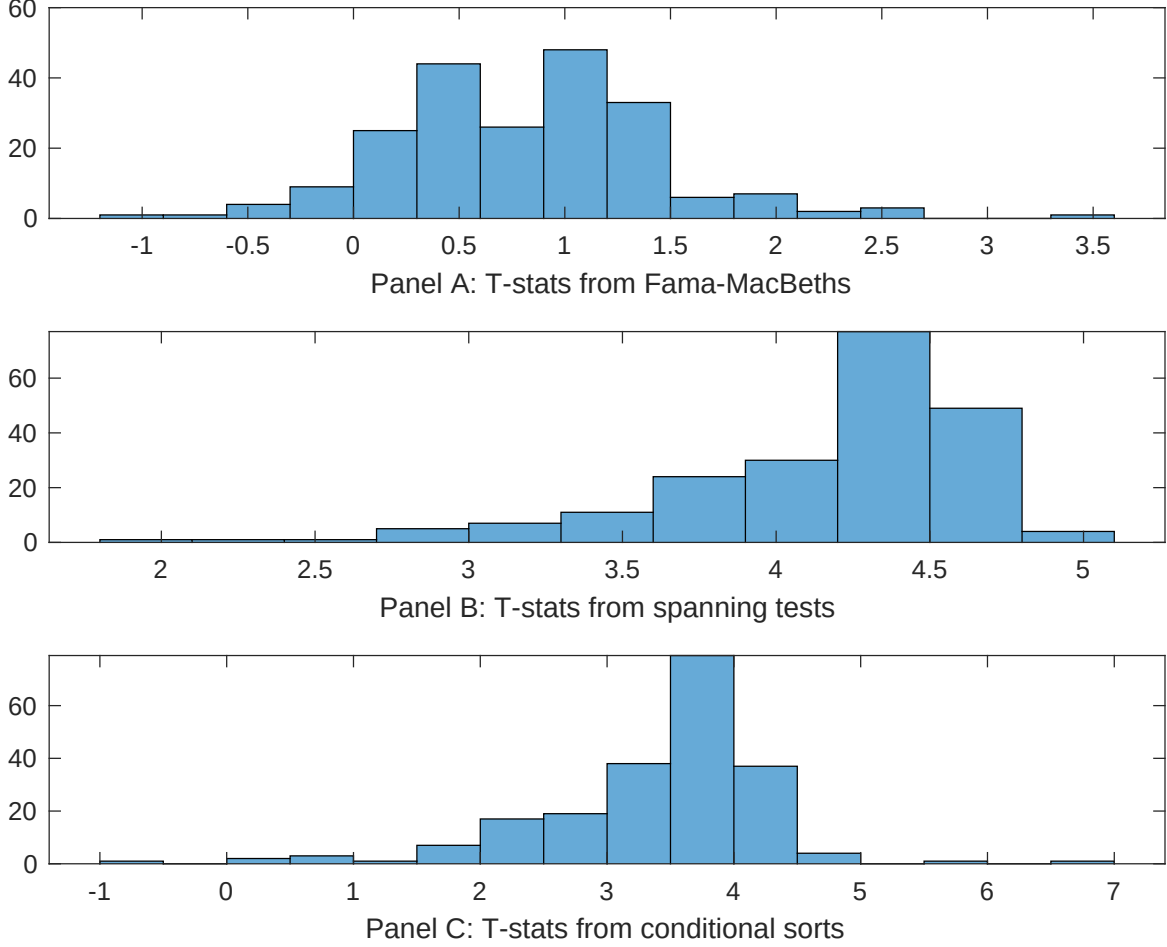


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CRD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRD}CRD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CRD. Stocks are finally grouped into five CRD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CRD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CRD}CRD_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.89]	0.17 [6.85]	0.13 [6.30]	0.12 [5.19]	0.12 [5.53]	0.13 [5.27]	0.12 [4.66]
CRD	0.19 [0.27]	0.99 [0.13]	-0.39 [-0.05]	0.38 [0.41]	0.35 [0.44]	0.62 [0.68]	0.22 [0.24]
Anomaly 1	0.15 [6.29]						0.79 [3.52]
Anomaly 2		0.43 [3.52]					-0.13 [-0.91]
Anomaly 3			0.28 [5.12]				0.27 [5.93]
Anomaly 4				0.23 [2.02]			0.13 [2.68]
Anomaly 5					0.12 [3.36]		0.13 [2.21]
Anomaly 6						0.18 [3.06]	-0.13 [-1.61]
# months	715	720	715	704	720	630	622
$\bar{R}^2(\%)$	0	0	0	1	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CRD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CRD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.19 [3.00]	0.20 [3.00]	0.13 [2.02]	0.24 [3.61]	0.21 [3.08]	0.22 [3.08]	0.16 [2.45]
Anomaly 1	32.53 [10.10]						15.11 [3.45]
Anomaly 2		21.20 [5.90]					20.38 [3.82]
Anomaly 3			31.99 [9.32]				21.63 [5.00]
Anomaly 4				15.75 [6.14]			12.14 [3.63]
Anomaly 5					6.82 [2.03]		-11.47 [-2.34]
Anomaly 6						11.19 [3.45]	-12.60 [-3.21]
mkt	-0.15 [-0.10]	-0.23 [-0.14]	0.49 [0.32]	1.51 [0.93]	-1.05 [-0.65]	1.34 [0.76]	1.48 [0.89]
smb	-0.29 [-0.13]	-2.25 [-0.98]	0.56 [0.25]	0.93 [0.40]	-1.63 [-0.70]	6.92 [2.47]	1.01 [0.38]
hml	-6.79 [-2.32]	-8.39 [-2.72]	-3.35 [-1.13]	-11.95 [-3.70]	-6.68 [-2.12]	-7.99 [-2.49]	-9.89 [-3.12]
rmw	-13.04 [-3.79]	5.60 [1.81]	-6.88 [-2.13]	-4.57 [-1.33]	5.22 [1.65]	3.30 [0.87]	-7.67 [-1.91]
cma	5.61 [1.19]	5.26 [0.92]	10.29 [2.22]	12.46 [2.52]	18.93 [3.32]	13.89 [2.67]	-8.01 [-1.35]
umd	3.13 [2.08]	4.27 [2.72]	2.53 [1.65]	5.88 [3.73]	4.82 [3.01]	4.37 [2.64]	2.98 [1.91]
# months	728	732	728	728	732	630	626
$\bar{R}^2(\%)$	19	12	18	12	8	8	22

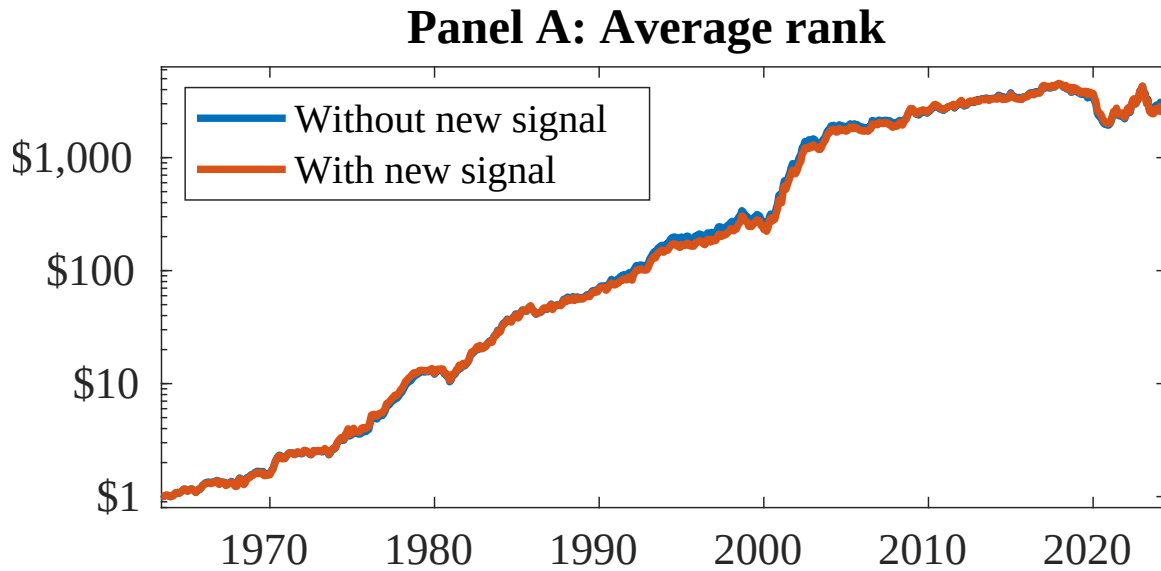


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CRD. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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